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Answer: C

Horizontal component of Normal contact force provides centripetal force

$$F_{net} = ma_c$$

$$N \sin \theta = \frac{mv^2}{r} \dots (1)$$

For vertical equilibrium: $N \cos \theta = mg \dots (2)$

$$\frac{(1)}{(2)}:$$

$$\tan \theta = \frac{v^2}{rg}$$

$$v = \sqrt{50 \times 9.81 \times \tan 25^\circ}$$

$$= 15.1 \text{ m s}^{-1}$$



